

Exercise 1

SQL Foundations: SELECT and Filtering

Selecting and Sorting

Table: Employees

id	first_name	last_name	department	Salary	hire_date	City
1	Alice	Green	IT	76000	2000-01-10	Johannesburg
2	Brian	Lee	HR	48000	2019-03-22	Cape town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokona	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q1 Retrieve all columns from the employees table

Answer: SQL

SELECT *
FROM employees;

id	first_name	last_name	department	Salary	hire_date	City
1	Alice	Green	IT	76000	2020-01-10	Johannesburg
2	Brian	Lee	HR	48000	2019-03-22	Cape town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokona	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

Q2 Find all unique departments.

Answer: SQL

SELECT DISTINCT department
FROM employees;

department
IT
HR
Finance
Marketing

Q3 Retrieve first and last names ordered by salary descending

Answer: SQL

SELECT first_name, last_name
FROM employees
ORDER BY Salary **DESC**;

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

Q4) Retrieve the top 3 highest-paid employees

Answer : SQL

```
SELECT id, first_name, last_name, Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;
```

id	first_name	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Emily	Truelove	65000

Q2 Filtering with WHERE

Q5 Find employees in the IT department

Answer: SQL

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';
```

id	first_name	last_name	department
1	Alice	Green	IT
5	Eva	Naïve	IT

Q6 Find employees in finance with salary > 60000

Answer: SQL

```
SELECT id, first_name, last_name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

id	first_name	last_name	department	salary
3	Cathy	Zula	Finance	65000

Q7) Find employees in HR or Marketing

Answer: SQL

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

id	first_name	last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

Q8) Find employees not in IT

Answer: SQL

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department NOT = 'IT';
```

```
SELECT id, first_name, last_name, department
FROM employees
WHERE NOT department = 'IT';
```

id	first_name	last_name	department
2	Brian	Lee	HR
3	Cathy	Zulca	Finance
4	David	Mokoena	Marketing

89) Find employees in IT, HR, or Finance using IN.

Answer: SQL

```
SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidu	IT

10) Find employees in IT with Salary > 65000 and city Johannesburg

Answer: SQL

```
SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidu	IT	72000	Johannesburg